ADEMP protocol: latent-class-mixture-application

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This document is the working protocol for the exploratory five-study simulation program described in `analysis/report/03-latent-class-mixture-application/simulation-plan.md`. The study is exploratory, and its findings should be confirmed in an independent study. ADEMP structure follows Morris, White and Crowther (2019). Each study states its Aims (A), Data-generating mechanisms (D), Estimands (E), Methods (M), and Performance measures (P).

Cells that share a value across studies are factored out to a common-infrastructure section to avoid redundant restatement.

Common infrastructure

Sample sizes. $N \in \{35, 70, 100\}$.

Trial designs. OL (open-label), CO (crossover), Hybrid, OL+BDC, generated by `pmsimstatsng::buildtrialdesign()`.

DGP-side carryover. $t_{1/2,\text{DGP}} \in \{0, 0.5, 1.0, 2.0\}$ weeks, with $\lambda_{\text{DGP}} = \ln 2 / t_{1/2,\text{DGP}}$ for $t_{1/2} > 0$ and $\lambda_{\text{DGP}} = \infty$ (immediate washout) for $t_{1/2} = 0$.

Analysis-side carryover. Where the analysis model is the linear-mixed baseline, $t_{1/2,\text{analysis}} \in \{0, 0.5, 1.0, 2.0\}$ varied independently of $t_{1/2,\text{DGP}}$ to inherit the analysis-side mis-specification axis.

Outcome model (linear mixed-effects baseline). Fixed analysis formula `Sx ~ bm + t + Dbc + bm:Dbc`, with continuous drug indicator $D_{it} = \exp(-\lambda_{\text{analysis}} t_{sd,it})$ off-drug and $D_{it} = 1$ on-drug. Random intercept `~1|ptID`, residual correlation `corCAR1(form = ~t|ptID)`.

Class-aware analysis. `lcmm::hlme` with two latent classes, biomarker entered in the class-membership submodel, fixed-effects structure matching the linear-mixed baseline, random intercept on the within-class model. Class-aware analyses store convergence status, log-likelihood, posterior class entropy, and modal class assignments alongside fitted coefficients.

Replicates. $n_{\text{reps}} = 1000$ per cell for power and bias estimands; $n_{\text{reps}} = 5000$ per cell for type I error and class-count recovery estimands.

Random-number control. Each replicate within a cell uses a deterministic per-replicate seed derived from the cell descriptor hash and a base seed fixed at study commencement. Base seed for the production runs is recorded in `01-base-seed.txt` at the time of run start.

Effect-size calibration. A cross-study calibration sub-study expresses each DGP's biomarker-treatment moderation parameter as a marginal Cohen's f^2-style standardised effect size at a reference time (week 8) and on-drug

state. The calibration sub-study runs once during Phase 1 and produces `calibration-table.rds`, which subsequent cells consult to align their effect-size axes.

Reference comparison. All cells include a $c_{bm} = 0$ (or class separation $\Delta = 0$) null sub-cell for type I error.

Study 1. Mixture-vs-MVN factorial

A. Decompose carryover-induced power loss for the biomarker- treatment interaction into approximation error in the DGP and inefficiency from a mis-specified analysis model.

D.

1. *Mixture DGP.* Two-class Bernoulli mixture, $Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i))$, with logistic gating $\pi(B_i) = \text{logit}^{-1}(\alpha + \beta B_i)$. Class prevalence at the biomarker mean fixed at 0.5 ($\alpha = 0$). Gating slope β chosen so that $\Pr(\hat{Z} = Z) = 0.75$ under modal posterior assignment. Class-conditional drug-response curves of modGompertz form with class-specific maxr: responder maxr^1 at the Hendrickson default; non-responder $\text{maxr}^0 = 0$. Other Gompertz parameters held at Hendrickson defaults. **PB-class correlation axis (`pb_class_correlation`).** Two cells: *PB-class-independent* ($\rho_{PB,Z} = 0$): PB drawn from the population distribution independently of class label, matching the original Hendrickson specification. *PB-class- correlated* ($\rho_{PB,Z} > 0$): responder class draws maxr^{PB} from a distribution shifted upward relative to the non-responder class, with shift calibrated so that $\text{Cor}(BR, PB)$ at on-drug timepoints equals a specified target (0.3 or 0.6). The PB-class-correlated cell is the natural mechanism for generating apparent subadditivity in a one-component additive analysis (see the subadditivity discussion in `analysis/report/06-component- decomposition/`); explicitly reporting power and bias under both cells lets the analyst distinguish 'true mechanistic subadditivity' from 'apparent subadditivity emerging from unmodelled latent-class structure'.
2. *MVN DGP* (covariance-only encoding, Architecture B as currently implemented).

Both DGPs are calibrated so that the marginal on-drug $\text{Cor}(B, BR_{it})$ at the calibration timepoint matches.

E.

- $\Delta_{\text{recovery}} = \pi^{\text{lcmm}}_{\text{mix}} - \pi^{\text{lme}}_{\text{mix}}$, where $\pi^{\text{anal}}_{\text{DGP}}$ denotes power for $\beta_{bm:D}$ at $\alpha = 0.05$.

- $\Delta_{\text{approx}} = \pi^{\text{lme}}_{\text{mix}} - \pi^{\text{lme}}_{\text{MVN}}$.
- Null check: $\pi^{\text{lcmm}}_{\text{MVN}} - \pi^{\text{lme}}_{\text{MVN}}$, which should not be reliably positive.

M. Linear-mixed baseline (`lme_analysis`) and class-aware analysis (`lcmm_analysis`, to be implemented in Phase 1) on each DGP, crossed with sample size, design, DGP carryover, and analysis carryover (the latter applies to the linear-mixed baseline only). Total cells: 3 DGP-cells $\times$ 2 anal $\times$ $3N \times 4$ design $\times\, 4t_{1/2,\text{DGP}} \times 4t_{1/2,\text{analysis}} = 1{,}152$, where the 3 DGP cells are the MVN reference, the PB-class-independent mixture, and the PB-class-correlated mixture.

P. Power, bias of $\hat{\beta}_{bm:D}$, Monte Carlo SE, nominal-95% coverage, convergence rate, fit time. Class-aware analyses additionally: posterior class entropy, modal class- assignment accuracy, BIC class-count selection.

Working prediction. $\Delta_{\text{recovery}} \geq 0$ asymptotically. Δ_{approx} sign indeterminate: if the mixture DGP under the linear-mixed analysis loses more or less power than the MVN DGP under the same analysis is the empirical question this study answers.

Study 2. Mean / covariance / combined moderation triptych

A. Verify that combined mean-plus-covariance moderation produces carryover-induced power loss intermediate between pure- mean and pure-covariance moderation, and quantify the interpolation.

D.

1. *Pure mean moderation* (Architecture A as implemented).
2. *Pure covariance moderation* (Architecture B as implemented).
3. *Combined Arminger-Stein-Wittenberg form.* Simultaneous mean scaling c^{mean}_{bm} and covariance moderation c^{cov}_{bm}. Mixing parameter $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$ interpolating from pure-mean ($\alpha = 0$) to pure-covariance ($\alpha = 1$); the calibration sub-study fixes the on-drug marginal $\text{Cor}(B, BR)$ constant across α.

E. Power for $\beta_{bm:D}$ as a function of α at each $t_{1/2}$. Working prediction: $\partial\pi/\partial\alpha < 0$ at $t_{1/2} > 0$ (combined and pure-covariance forms lose more power than pure-mean as carryover increases) and $\partial\pi/\partial\alpha \approx 0$ at $t_{1/2} = 0$ (no carryover means no differential erosion).

M. Linear-mixed baseline only. Total cells: $5\alpha \times 3N \times 4$ design $\times\, 4t_{1/2,\text{DGP}} \times 4t_{1/2,\text{analysis}} = 960$.

P. Power, bias, Monte Carlo SE, coverage, convergence.

Study 3. Heterogeneous random-slopes DGP

A. Test the Verbeke-Lesaffre / Proust-Lima heterogeneous- random-effects representation as a more biologically transparent alternative to covariance-only moderation, holding the analysis model at the linear-mixed baseline.

D.

1. *Heterogeneous random slopes.* Two-component mixture over the random-slope distribution on the drug indicator: responder component $\beta_R \sim \mathcal{N}(\mu_R, \sigma^2_R)$, non-responder $\beta_{NR} \sim \mathcal{N}(0, \sigma^2_{NR})$. Mixing probability $\pi(B_i)$ logistic in B_i, with the same gating-slope calibration as Study 1.
2. *Covariance-only baseline* at matched marginal $\mathrm{Cor}(B, BR)$ from the calibration sub-study.

E. Power, calibrated to the matched-effect-size axis; identifiability metrics for `lcmm` under the random-slopes DGP at small N.

M. Linear-mixed baseline and `lcmm_analysis`. Total cells: 2 DGP $\times$ 2 anal $\times$ $3N \times 4$ design $\times\ 4t_{1/2,\mathrm{DGP}} \times 4t_{1/2,\ \mathrm{analysis}} = 768$.

P. Power, convergence rate, posterior class entropy.

Study 4. Three-regime breakdown grid

A. Map the regions of parameter space where the MVN approximation to a latent-class DGP fails, and quantify failure.

D. Three latent-class DGPs, each varying one regime axis with the others held at a 'mild' baseline.

1. *Class-separation axis.* Vary the responder/non-responder maxr gap Δ over five values from 0 to twice the Hendrickson default. At small Δ the marginal on-drug BR histogram is unimodal; at large Δ it is bimodal. Operationalising metric: bimodality coefficient.
2. *Gating-slope axis.* Vary the logistic slope of $\pi(B_i)$ over five values from 'gradual' (slope at zero corresponding to $\Pr(\hat{Z} = Z) = 0.6$ under modal assignment) to 'near-step' ($\Pr(\hat{Z} = Z) = 0.95$). A thresholded mean-moderation variant of Architecture A is the natural comparator at the steep end and is implemented as part of Phase 1 infrastructure.
3. *Class-varying covariance axis.* Vary the ratio σ^2_R/σ^2_{NR} over five values centred on 1, with class-conditional means held equal. Within-class AR(1) parameter ρ varied jointly when justified by clinical plausibility.

E. For each regime, the location at which the linear-mixed analysis loses more than 5% power relative to the matched mixture analysis; and the location at

which the mixture analysis fails identifiability checks (posterior entropy below 0.7 or convergence rate below 90%).

M. Linear-mixed baseline and `lcmm_analysis`. Three one- dimensional sweeps (not a full three-way factorial), each crossed with sample size and DGP carryover. Total cells: 3 axis $\times$ 5 axis-position $\times$ 2 anal $\times$ $3N \times 4t_{1/2,\text{DGP}} = 360$, optionally crossed with design (multiplying by 4 if budget permits).

P. Power, bimodality coefficient (DGP 1 only), σ ratio recovery (DGP 3 only), convergence, entropy.

Study 5. Small-N identifiability and type I error

A. Establish whether mixture-model analyses are usable at N-of-1-relevant sample sizes, independent of any specific moderation question. This study is a prerequisite for substantive interpretation of Studies 1, 3, and the lcmm arm of Study 4.

D.

1. *Single-component, gaussian.* Covariance-only encoding with $c_{bm} = 0$.
2. *Single-component, mildly non-gaussian.* Heavy-tailed and mildly skewed residual distributions on BR: t_5 distributed standardised residuals, and (separately) skew-normal with shape parameter 4.
3. *True two-class.* Mixture DGP from Study 1, varied across class separation $\Delta \in \{0, 0.5, 1.0, 1.5, 2.0\}\times$ Hendrickson-default.

E.

- Type I error for $\beta_{bm:D}$ under cell 1, target 0.05.
- Class-count selection frequencies (BIC and BLRT) under cells 2 and 3, with $K \in \{1, 2, 3\}$ entered into model comparison.
- Posterior entropy distribution under cell 3 as a function of N and Δ.
- Convergence rate and mean fit time for each cell.

M. `lcmm::hlme` with $K \in \{1, 2, 3\}$ classes for cells 2 and 3; `lcmm::hlme` with $K = 2$ for cell 1 (the type-I-error sub- cell). Total cells: 3 DGP-family $\times$ 5 DGP- parameter $\times$ $3N \times 3K = 135$ for cells 2 and 3; $1 \times 1 \times 3N \times 1 = 3$ for cell 1.

P. Type I error, class-count selection rate, convergence rate, posterior entropy, fit time.

Phase 1 deliverables and sequencing

The protocol above specifies cell counts: roughly 2300 cells across the five studies, each with 1000-5000 replicates. A pilot at one cell per study during Phase 1 characterises per-cell compute time and bounds the production- run budget.

Phase 1 infrastructure deliverables, in approximate dependency order:

1. `lcmm_analysis()` wrapper around `lcmm::hlme` with the same input/output contract as `lme_analysis()`.
2. Heterogeneous-random-slopes DGP code path (Study 3).
3. Marginal-mixture DGP code path (Study 1).
4. Combined-moderation DGP code path (Study 2).
5. Thresholded mean-moderation variant (Study 4 gating-slope axis).
6. Effect-size calibration sub-study script.
7. Pilot driver running one cell per study with timing.

Phase 2 (Study 5 pre-flight) follows; Phase 3 (Studies 1, 2, 3) runs in parallel; Phase 4 (Study 4) closes the programme.

Changes to the working protocol

Any change to the cells, estimands, methods, or performance measures specified in this working protocol is recorded in `02-deviations-log.md` with date, reason, and reviewer signoff (if external review is sought).

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